

Unit 3: The Precise Definition of a Limit

1. The idea, in plain words

Back in Unit 2, we said a limit is “the number $f(x)$ gets closer and closer to as x gets closer and closer to a .” That’s a great starting picture, but “closer and closer” is a little fuzzy — closer by how much? This unit just makes that fuzzy idea precise, using a simple game: the tolerance game.

Picture it like archery, or a factory quality check. Suppose someone hands you a target number L and says: “I want $f(x)$ to land within a tiny margin of L — say, within ε (the Greek letter epsilon) — no matter how small a margin I ask for.” Your job is to find a matching rule for x : “As long as x stays within some margin δ (the Greek letter delta) of a , I guarantee $f(x)$ will land inside your target band.”

- ε (epsilon) = how close you want the output, $f(x)$, to be to L . This lives on the y-axis (the output side).
- δ (delta) = how close the input, x , needs to be to a to guarantee that. This lives on the x-axis (the input side).

The picture: draw two horizontal dashed lines at $L - \varepsilon$ and $L + \varepsilon$ — that’s the “target band” for the output. Then find two vertical dashed lines at $a - \delta$ and $a + \delta$ — the “safe zone” for the input — such that any x inside that safe zone (other than $x = a$ itself) produces an $f(x)$ that lands inside the target band.

Saying “the limit is L ” really just means: no matter how thin someone makes that target band (ε), you can always find a safe zone (δ) around a that keeps you inside it. That’s all the “precise definition” is doing — turning “gets closer and closer” into an actual tolerance-matching game you can play with real numbers.

Good news for this unit: we’ll only play this game with straight-line functions ($f(x) = mx + b$), because for a line, there’s a clean, simple shortcut for finding a matching δ — no formal proof-writing needed, just some algebra.

2. Toolbox

What each symbol means:

- a = the input value we’re approaching
- L = the limit (the target output value), often $L = f(a)$ for our examples
- ε = the allowed margin of error on the output (how close $f(x)$ must be to L)
- δ = the allowed margin of error on the input (how close x must be to a)

The tolerance-matching statement: we want to guarantee

$$|f(x) - L| < \varepsilon \quad \text{whenever} \quad 0 < |x - a| < \delta$$

The straight-line shortcut: if $f(x) = mx + b$ and $L = f(a)$, then

$$|f(x) - L| = |m(x - a)| = |m| \cdot |x - a|$$

So $|f(x) - L| < \varepsilon$ exactly when $|x - a| < \frac{\varepsilon}{|m|}$. That means:

$$\delta = \frac{\varepsilon}{|m|}$$

always works. (Any smaller delta than this also works — it’s not the one “correct” answer, just a safe one.)

Working backward: if you’re told a δ was used, the guaranteed output tolerance is

$$\varepsilon = |m| \cdot \delta$$

3. Common mistakes

- Mixing up which axis epsilon and delta belong to. Epsilon is always about the output (y), delta is always about the input (x).
- Thinking there’s only one “correct” delta. Any delta smaller than or equal to $\varepsilon/|m|$ works. If a smaller window on x works, a slightly larger one that’s still $\leq \varepsilon/|m|$ works too — there isn’t a single magic number.
- Forgetting to use the absolute value / size of the slope. A steep negative slope (like $m = -5$) still needs $|m| = 5$ in the formula — the sign of the slope doesn’t matter here, only its size.
- Assuming a bigger epsilon needs a bigger delta — that part’s true — but forgetting that a steeper line needs a smaller delta for the same epsilon. A steep line amplifies small changes in x into big changes in y , so you need to control x more tightly.
- Forgetting to double-check by plugging the boundary values back in. Plugging in $x = a + \delta$ and $x = a - \delta$ is a great way to confirm your delta actually lands $f(x)$ right at (or inside) the tolerance band.

4. Problem Set

Warm-up

1. For $f(x) = 2x$ at $a = 3$ (so $L = 6$): if we want $f(x)$ to stay within $\varepsilon = 0.4$ of 6, how close must x be to 3? Find a valid δ .
2. For $f(x) = 5x$ at $a = 2$ (so $L = 10$), with $\varepsilon = 1$: find a valid δ .
3. For $f(x) = -3x$ at $a = 1$ (so $L = -3$), with $\varepsilon = 0.6$: find a valid δ .
4. For $f(x) = x + 4$ at $a = 5$ (so $L = 9$), with $\varepsilon = 0.3$: find a valid δ .
5. In plain words: if two lines have the same ε requirement, but one line is steeper (larger $|m|$) than the other, which one needs the smaller δ ? Explain why, using the “amplifying” idea from the Common Mistakes section.
6. In plain words: if someone tightens their tolerance request (makes ε smaller), does the matching δ need to get smaller too, stay the same, or get bigger? Explain.

Standard

7. For $f(x) = 4x - 1$ at $a = 2$ (so $L = 7$), with $\varepsilon = 0.8$: find a valid δ .
8. For $f(x) = -2x + 5$ at $a = 3$ (so $L = -1$), with $\varepsilon = 0.5$: find a valid δ .
9. For $f(x) = 6x + 2$ at $a = -1$ (so $L = -4$), with $\varepsilon = 1.2$: find a valid δ .
10. For $f(x) = 3x$ at $a = 0$, with $\varepsilon = 0.09$: find a valid δ .
11. For $f(x) = 2x + 1$ at $a = 4$: if you use $\delta = 0.15$, what is the resulting ε this guarantees? Describe the output tolerance band in words (i.e., “ $f(x)$ stays between ____ and ____”).

12. For $f(x) = -4x + 3$ at $a = 1$: if you use $\delta = 0.1$, what is the resulting ε ? Describe the output tolerance band in words.

Challenge

13. For $f(x) = 7x - 3$ at $a = 2$, with $\varepsilon = 0.35$: find the delta given by the shortcut formula, then check your answer by plugging $x = a + \delta$ and $x = a - \delta$ into $f(x)$ and confirming both land exactly on the edge of the tolerance band around L .
14. For $f(x) = 3x + 1$ at $a = 2$ (so $L = 7$), with $\varepsilon = 0.6$: the shortcut formula gives one delta. But suppose there's also a separate physical requirement that x must stay within 0.15 units of $a = 2$ no matter what. What overall δ should you use, and why?
15. For $f(x) = -5x + 2$ at $a = -2$ (so $L = 12$), with $\varepsilon = 0.25$: find the delta given by the shortcut formula. Then, is $\delta = 0.02$ also a valid choice? Explain why or why not, using the “smaller delta” idea from the Toolbox.
16. Suppose you're told that for $f(x) = 2x$ near $a = 3$, the output $f(x)$ must stay strictly between 5.7 and 6.3. Figure out what ε was being used, and then find the matching δ .

Applied

17. A machine cuts metal rods, and the rod length (in meters) depends on a dial setting x according to $f(x) = 2x$. The target length is 6 meters (at dial setting $a = 3$), and a rod is acceptable only if it's within 0.02 m of 6 m. How precisely must the dial be set (find δ) to guarantee an acceptable rod?
18. A thermostat's resulting room temperature (in $^{\circ}\text{F}$) is $T(x) = 3x + 65$, where x is a dial setting. The target temperature is 71°F at dial setting $a = 2$. The room must stay within 0.9°F of the target. How precisely must the dial be set?
19. A pharmacist mixes a solution whose concentration (in mg/mL) is $C(x) = 4x + 10$, where x is the amount (in mL) of a stock solution added. The target concentration is 26 mg/mL at $x = 4$ mL. Safety rules require the concentration to stay within 0.5 mg/mL of the target. How precisely must x be measured?
20. A drone's altitude (in meters) is modeled by $h(x) = 10x$, where x is a throttle setting near $a = 5$ (giving a target altitude of $L = 50$ m). To stay clear of a restricted zone, the altitude must stay within 2 m of 50 m. How precisely must the throttle setting be controlled?